Natural Language Processing and Attention Mechanisms

Natural language processing involves enabling computers to understand and process human languag

Traditional information retrieval relies heavily on bag-of-words representations and TF-IDF term weigh

TF-IDF penalizes common words that appear across all documents while rewarding terms distinctive t

Cosine similarity is widely used in vector space models to compute the orientation angle between docu

Stemming and lemmatization reduce words to their base linguistic root forms, improving term recall.

Page 2: Modern Transformer Architectures

The Transformer architecture introduced self-attention mechanisms that process input tokens in parall

Unlike recurrent neural networks, transformers avoid sequential bottlenecks, enabling massive scalabi

BERT utilizes bidirectional encoder representations to capture deep contextual semantics in text.

Dense embeddings from sentence transformers map passages to continuous vector spaces based on

Retrieval-Augmented Generation (RAG) combines semantic document search with large language mo